---

## 📘 PART A: CONCEPTUAL FOUNDATION (COMPLETE)

---

### ◈ 1. What is Data Analysis?

**Answer:**

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to extract meaningful insights and support decision-making.

It involves:

- Data collection
- Data cleaning
- Exploratory analysis
- Visualization
- Interpretation

👉 **Goal:** Convert raw data into useful information.

---

### ◈ 2. How to Plan a Data Science Project

**Answer:**

A data science project follows a structured approach:

**Steps:**

1. **Problem Definition**
    - Understand business objective
2. **Data Collection**
    - Gather data from multiple sources
3. **Data Cleaning**
    - Handle missing values, duplicates, outliers
4. **Exploratory Data Analysis (EDA)**
    - Understand patterns and relationships
5. **Feature Engineering**
    - Create meaningful features
6. **Model Building**
    - Select and train ML models
7. **Evaluation**

    - Measure performance using metrics
8. **Deployment**
    - Use model in real-world application

---

### ◈ 3. How to Frame a Machine Learning Problem

**Answer:**

Framing an ML problem means converting a real-world problem into a machine learning task.

**Steps:**

- Identify **target variable**
- Define **input features**
- Decide **type of problem**:
    - Classification
    - Regression
- Choose **evaluation metric**

👉 Example (your project):

- Problem: Loan Default Prediction
- Type: **Classification**
- Target: default_flag

---

### ◈ 4. Tensors (In-Depth Explanation)

---

### 📌 What is a Tensor?

A **tensor** is a multi-dimensional array used to store numerical data.

| Type | Example |
|---|---|
| Scalar (0D) | Single value |
| Vector (1D) | List |
| Matrix (2D) | Table |
| Tensor (3D+) | Multi-dimensional data |

---

### 📌 Tensor Types with NumPy

```
import numpy as np

# 0D Tensor (Scalar)
scalar = np.array(10)
print("Scalar:", scalar)

# 1D Tensor (Vector)
vector = np.array([1, 2, 3])
print("Vector:", vector)

# 2D Tensor (Matrix)
matrix = np.array([[1, 2], [3, 4]])
print("Matrix:\n", matrix)

# 3D Tensor
tensor_3d = np.array([
    [[1, 2], [3, 4]],
    [[5, 6], [7, 8]]
])
print("3D Tensor:\n", tensor_3d)
```

---

### 📌 Tensor Properties

```
print("Shape:", tensor_3d.shape)   # Dimensions
print("Size:", tensor_3d.size)     # Total elements
print("Dimensions:", tensor_3d.ndim)  # Number of axes
```

---

### 📌 Basic Tensor Operations

```
a = np.array([1,2,3])
b = np.array([4,5,6])

# Addition
print(a + b)
```

```
# Multiplication
print(a * b)

# Dot Product
print(np.dot(a, b))
```

---

📌 **Why Tensors are Important**

- Used in **Machine Learning & Deep Learning**
- Efficient for numerical computation
- Foundation of libraries like:
  - TensorFlow
  - PyTorch

---